PROBLEM STATEMENT FOR LAB BOOKING SYSTEM

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1. HIGH LEVEL PROBLEM SUMMARY

- **Elevator Statement**

  I am creating a Laboratory Booking System to help manage lab schedules in our university. Instructors can easily book labs, and the system checks if the lab and equipment are available. It keeps track of lab use, equipment, and bookings. Students, instructors, and lab staff can use it. This system helps avoid booking problems and saves time by making everything clear and organized.

- **Primary Success Criteria**
  - ✓ Let instructors book labs only if they are available.
  - ✓ Keep correct records of all lab bookings and equipment status.
  - ✓ Create reports about lab usage to help with planning and analysis.
  - ✓ Make the system easy to use for students, instructors, and lab staff.

- **Scope of the Project**

  What is included:

  - A system to book university labs based on availability.
  - User roles like students, instructors, lecturers in charge, and lab technical officers.
  - A lab database with details like lab ID, type, capacity, and equipment.
  - The ability for instructors to request bookings and get confirmations.
  - Checking availability of labs and equipment before confirming a booking.
  - Keeping records of lab bookings and equipment use.
  - Generating reports on lab usage for analysis.
  - A simple and friendly interface for all users.

  What is not included:

  - Real-time equipment tracking with sensors or IoT devices.
  - Mobile app development (only web-based or desktop system for now).
  - Maintenance or repair system for lab equipment.
  - Automatic approval by AI (approval will be basic and rule-based).

2. DETAILED PROBLEM STATEMENT.

2.1 Function

The system should enable:

- Allow instructors to request lab bookings.
- If there is a lab booking request. The lab technical officer will check the labs and before approving the reservation request.
- Store lab details like lab ID, type, capacity, and availability.
- Records the date the lab was booked, whether it was completed or not, and the name of the lab.
- Students, lecture in charge can only view the lab schedule
- Generate reports for lab usage to help with planning.
- Be easy to use for all users—students, instructors, lab staff, and lecturers.

2.2Time

Past

Many labs are booked using manual methods, such as paper forms or emails. This can cause confusion, double-booking, and missing records.

Present

With this system, instructors and staff can use a digital platform to book labs and manage equipment more easily and clearly. Everyone can see real-time updates and avoid scheduling issues.

Future

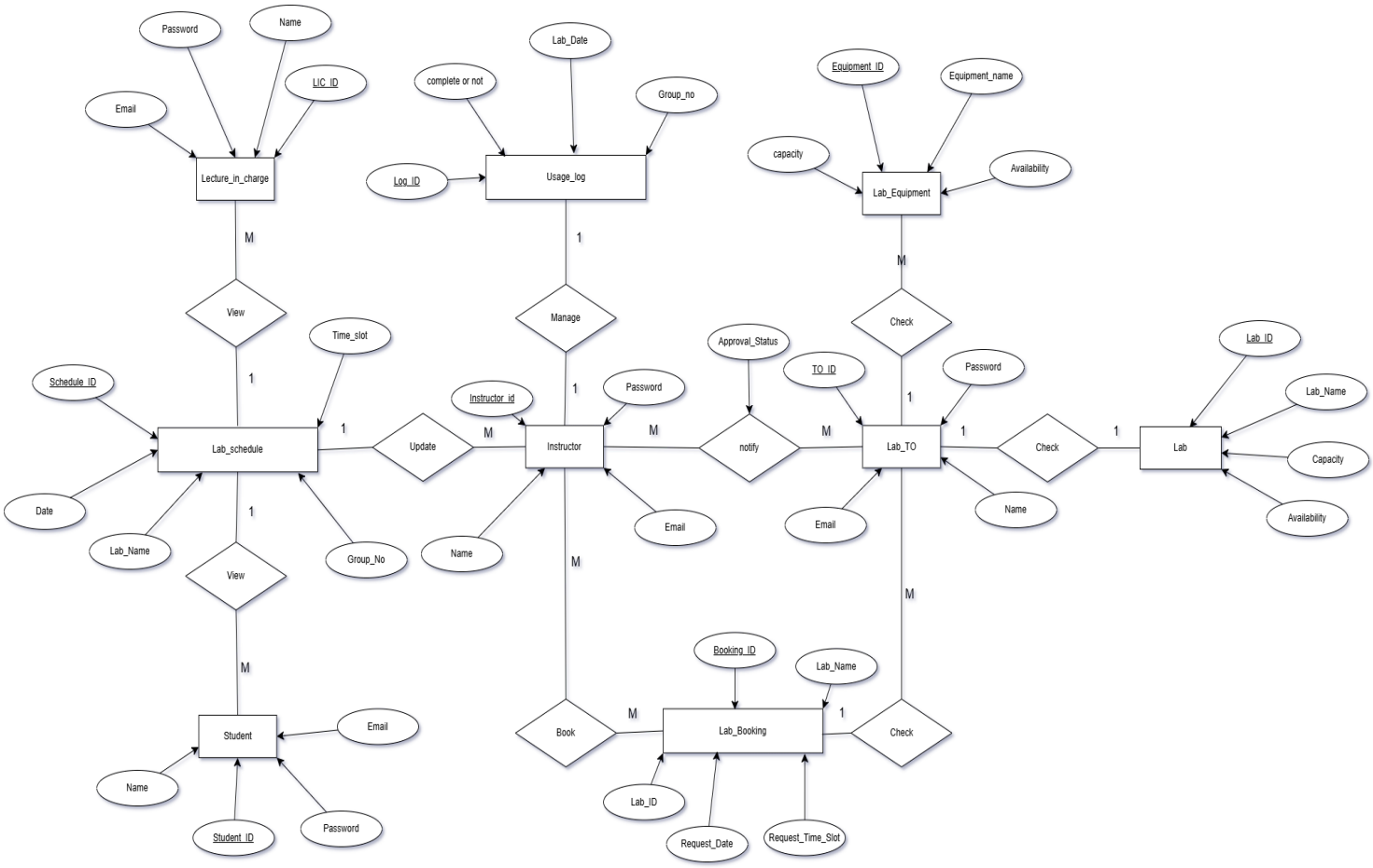
In the future, the system can be improved to include:

- A mobile app version.
- Automatic notifications/reminders for bookings.
- Integration with equipment maintenance tracking.
- More advanced analytics and dashboards.

3. KEY STAKEHOLDERS.

Name	Role
Student	View the lab schedule
Lecture in charge	View the lab schedule
Lab Instructor	Booking a lab, update lab schedule and Manage the usage log
Lab technical officer	Checks laboratory availability, capacity, and required equipment before approving the booking request

4. REVISION HISTORY.

Date	Version	Reason for change	Author
2025-06-03	1.0	Initial proposal submission	Wickramasinghe G.A.G.M.

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